Linear Regression Models

Linear regression models are a cornerstone of statistical and machine learning techniques, offering a simple yet powerful way to model and analyze relationships between variables. These models are widely used in fields like data analysis, predictive modeling and artificial intelligence.

What is Linear Regression?

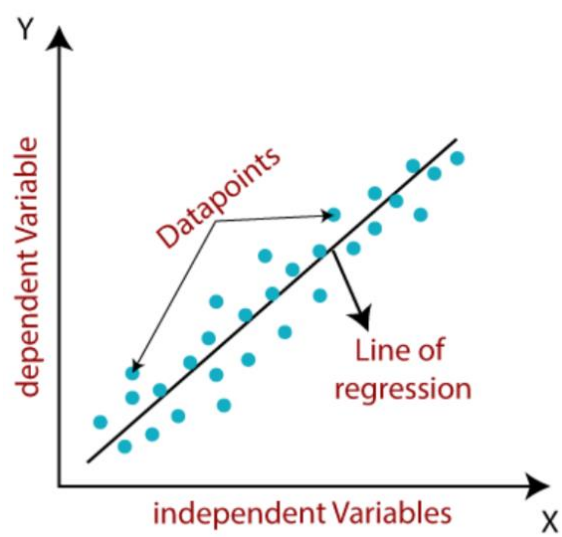
Linear regression is a statistical technique used to establish a relationship between a dependent variable (also called the target or output) and one or more independent variables (also called features or predictors). The goal is to find the best-fitting straight line that minimizes the difference between predicted and actual values.

Mathematical Representation

The outcome of a process is often represented by a variable y called the dependent or study variable, which depends on k independent variables $X_1, X_2 \dots \dots X_k$. The relationship between y and these variables can be written as:

$$y = f(X_1,\ X_2 \dots \dots X_k, \beta_1,\ \beta_2. \dots \beta_k) + \varepsilon$$

Here, f is a function that explains how the independent variables $X_1,\ X_2 \dots \dots X_k$ affect y. The values $\beta_1,\ \beta_2. \dots \beta_k$ are parameters that show how much each variable contributes. The term ε represents random variation (error). A model or relationship is termed as linear if it is linear in parameters and nonlinear if it is not linear in parameters. The linear regression model provides a sloped straight line representing the relationship between the variables.



Best fit line:

When working with linear regression, our main goal is to find the best fit line that means the error between predicted values and actual values should be minimized. The best fit line will have the least error.

Types of Linear Regression: Linear regression can be further divided into two types

1. Simple linear regression
2. Multiple linear regression

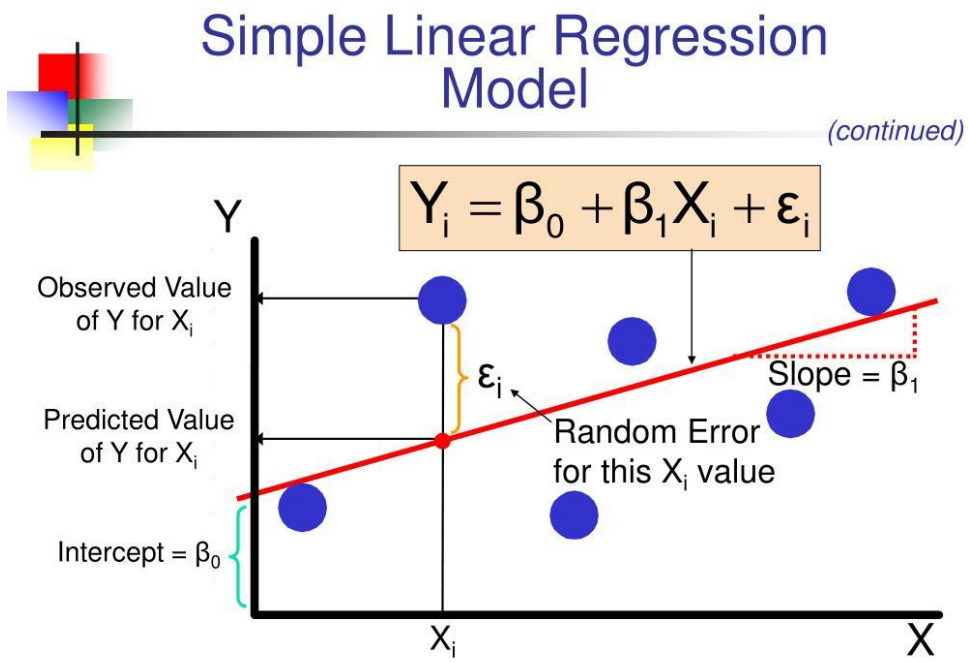
Simple linear regression models:

A Simple linear regression model is a statistical technique used to model the relationship between a single dependent variable Y and a single independent variable X. It assumes a linear relationship between the predictor X and the response Y, which can be represented by a straight line in a two-dimensional space.

The simple linear regression model is represented by the equation:

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

- Y is the dependent variable (response).
- X is the independent variable (predictor).
- β_0 is the intercept. (the value of Y when $X = 0$).
- β_1 is the slope (the change in Y for a unit change in X).
- ϵ is the error term, representing the variability in Y that cannot be explained by X.



Assumptions

1. Linearity

Linearity assumes a linear relationship between the dependent variable Y and all independent variables X_i.

2. Independence of Errors

The second assumption is that errors or residuals are independent of each other that are uncorrelated.

3. Homoscedasticity

Homoscedasticity implies that the variance of the errors should be constant.

4. Normality of Errors

It assumes the ε are normally distributed.

Estimated coefficients of Simple linear regression model:

Consider the data points $(x_i, y_i); i = 1, 2, \dots n$. Let $y = \beta_0 + \beta_1 x + \varepsilon$ be the line of best fit for the given data where ε is error term.

For each data point

$$y_i = \beta_0 + \beta_1 x_i \qquad (1)$$

$\Rightarrow \sum_{i=1}^{n} \mathrm{y_i} = n\,\beta_0 + \beta_1 \sum_{i=1}^{n} x_i$

$\Rightarrow \frac{1}{n}\sum_{i=1}^{n} y_i = \beta_0 + \beta_1 \frac{1}{n}\sum_{i=1}^{n} x_i$

$\Rightarrow \quad \overline{y} = \beta_0 + \beta_1 \overline{x}$ (2)

Also, from equation (1)

$x_i y_i = \beta_0 x_i + \beta_1 x_i^2$

$\Rightarrow \sum_{i=1}^{n} x_i y_i = \beta_0 \sum_{i=1}^{n} x_i + \beta_1 \sum_{i=1}^{n} x_i^2$ (3)

By shifting the origin to $(\overline{x}, \overline{y})$ we get,

$\sum_{i=1}^{n}(x - \overline{x})\,(y - \overline{y}) = \beta_0 \sum_{i=1}^{n}(x_i - \overline{x}) + \beta_1 \sum_{i=1}^{n}(x - \overline{x})^2$

But $\sum_{i=1}^{n}(x_i - \overline{x}) = \sum_{i=1}^{n} x_i - \sum_{i=1}^{n} \overline{x}$

$= n\overline{x} - n\overline{x} = 0$

Therefore,

$$\sum_{i=1}^{n}(x - \overline{x})\,(y - \overline{y}) = \beta_1 \sum_{i=1}^{n}(x - \overline{x})^2$$

$$\Rightarrow \hat{\beta}_1 = \frac{\sum_{i=1}^{n}(x - \overline{x})\,(y - \overline{y})}{\sum_{i=1}^{n}(x - \overline{x})^2}$$

and

$\hat{\beta}_0 = \overline{y} + \hat{\beta}_1 \overline{x}$

Multiple linear Regression models:

Multiple Linear Regression (MLR) is a statistical method used to model the relationship between one dependent variable (target variable) and two or more independent variables (predictors). It extends simple linear regression, which deals with one independent variable, to analyze and predict outcomes based on multiple factors.

Why Use MLR?

1. Prediction: To predict the value of the dependent variable based on the independent variables.

2. Relationship Analysis: To understand how each predictor influences the dependent variable.

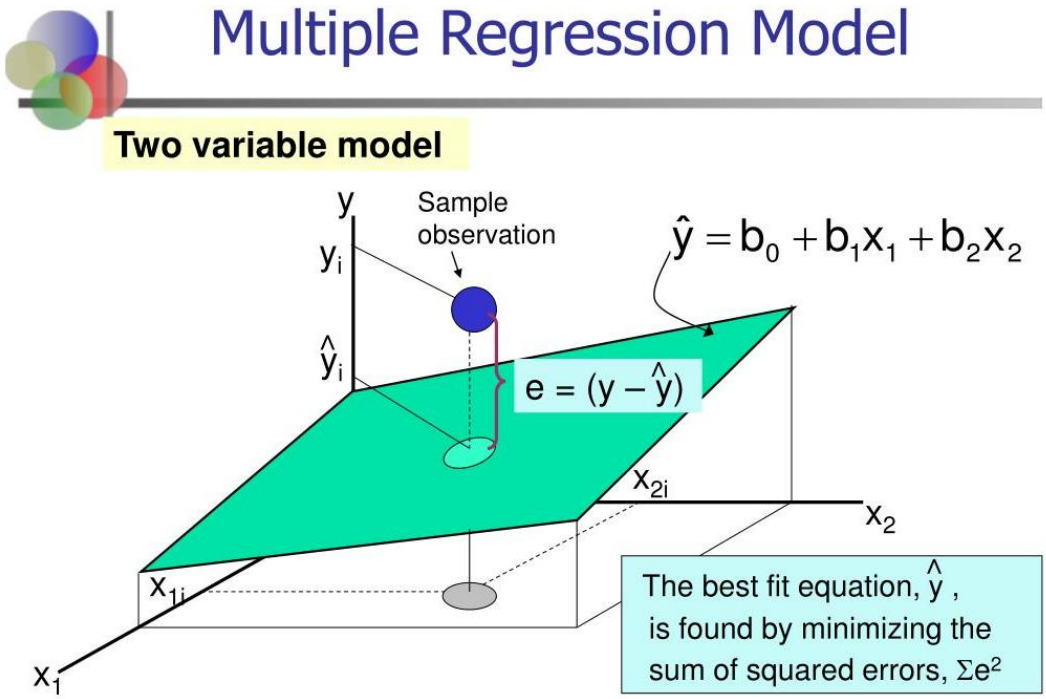
3. Decision-Making: Helps in optimizing processes by understanding key influencing factors.

Equation of MLR

The general equation for MLR is:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_n X_n + \epsilon$$

- Y: Dependent variable (e.g., sales, profit).
- $X_1, X_2 \ldots . X_n$: Independent variables (e.g., price, marketing budget, location).
- β_0: Intercept
- $\beta_1, \beta_2, \ldots \beta_n$: Regression coefficients
- ϵ: Error term



Interpretation of Multiple Linear Regression coefficients:

1. Intercept (β_0): Represents the baseline level of the dependent variable when none of the predictors contribute to the outcome.

Example: If, $Y = 50 + 5X_1 + 10X_2$, the intercept (50) means that Y will be zero when $X_1 = 0$ and $X_2 = 0$.

2. Regression coefficients ($\beta_1, \beta_2, \ldots \beta_n$): Each $\beta_i$represents the expected change in Y for a one-unit increase in X_i holding all other variables constant.

Interpretation:

1. A positive coefficient ($\beta_i > 0$) indicates that as X_i increases, Y also increases.

2. A negative coefficient ($\beta_i < 0$) indicates that as X_i increases, Y decreases.

Assumptions of MLR

1. Linearity: The relationship between dependent and independent variables is linear.
2. Independence: Observations are independent of each other.
3. Homoscedasticity: The variance of errors is constant across all levels of X.
4. Normality: Errors are normally distributed.

Evaluation Metrics for Multiple Linear Regression (MLR)

1) R-Squared (R^2): Measures the proportion of the total variation in the dependent variable (Y) explained by the independent variables $X_1, X_2 \ldots . X_n$.

   Formula: $R^2 = \frac{SS_{reg}}{SS_{Total}}$

where:

- SS_{reg}: Regression sum of squares (variation explained by the model).

$$SS_{reg} = \beta_1 . \sum (x - \overline{x})(y - \overline{y})$$

- SS_{Total}: Total sum of squares (total variation in Y).

$$SS_{Total} = \sum (y - \overline{y})^2$$

2) Residual sum of squares (SS_{res}): Measures the total variation in the dependent variable Y that is not explained by the regression model.

$$SS_{res} = SS_{Total} - SS_{reg}$$

3) **F-Statistic**: Tests the overall significance of the model. Higher F-statistic indicates that the model is statistically significant.

Applications in Computer Science:

- Performance Optimization: Analyze factors affecting system performance.
- Software Cost Estimation: Predict the cost of a software project based on team size, duration and complexity.
- Predictive Modeling: Forecast user behavior, like app usage based on age, location and device type.

Examples:

1) Employee Salary Prediction

   Problem: Predict an employee's annual salary Y based on:

   X_1: Years of experience

   X_2: Education level (e.g., bachelor's, master's, PhD)

   X_3: Performance rating

   Model:

   Salary$= 30000 + 5000(Experience) + 10000(Education) + 2000(Rating) + \varepsilon$

2) **Software Project Cost Estimation**

   Problem: Predict the cost of a software project (Y) based on:

   X_1: Team size

   X_2: Development time (in months)

   X_3: Project complexity (rated on a scale of 1 to 10)

   Model:

   $$Cost = 10000 + 2000(Team\ size) + 3000(Time) + 5000(Complexity) + \varepsilon$$

<u>Problems based on Simple linear regression and MLR model</u>

Q.1. Calculate the line of regression for the following data:

x	1	2	3	4
y	2	4	5	7

Solution: Calculate means of x and y

$\overline{x} = 2.5$ and $\overline{y} = 4.5$

x	y	$(x-\overline{x})$	$(y-\overline{y})$	$(x-\overline{x})(y-\overline{y})$	$(x-\overline{x})^2$
1	2	-1.5	-2.5	3.75	2.25
2	4	-0.5	-0.5	0.25	0.25
3	5	0.5	0.5	0.25	0.25
4	7	1.5	2.5	3.75	2.25
				$\sum(x-\overline{x})(y-\overline{y})=8$	$\sum(x-\overline{x})^2=5$

Calculate slope:

$$\beta_1 = \frac{8}{5} = 1.6$$

Calculate intercept:

$$\beta_0 = 4.5 - 1.6(2.5) = 0.5$$

Regression line: $$y = 1.6x + 0.5$$

Q.2. The yield y of a chemical process is a random variable whose value is considered to be a linear function of the temperature x. The following data of corresponding values of x and y is found:

Temperature (x)	0	25	50	75	100
Yield (y)	14	38	54	76	95

Calculate the line of regression.

Q.3. The following data relate to advertisement expenditure (in lakh of rupees) and their corresponding sales (in crores of rupees):

Advertisement expenditure (x)	40	50	38	60	65	50	35
Sales (y)	38	60	55	70	60	48	30

Estimate the sales corresponding to advertising expenditure of Rs. 30 lakhs.

Solution:

Step 1: Organize the Data in a Table

x	y	$(x-\overline{x})$	$(y-\overline{y})$	$(x-\overline{x})(y-\overline{y})$	$(x-\overline{x})^2$

40	38	-5.71	-15.14	86.57	32.64
50	60	4.29	6.86	29.44	18.41
38	55	-7.71	1.86	-14.34	59.48
60	70	14.29	16.86	240.88	204.12
65	60	19.29	6.86	132.26	372.11
50	48	4.29	-5.14	-22.06	18.41
35	30	-10.71	-23.14	247.89	114.71
$\sum x = 338$	$\sum y = 361$			$\sum(x-\overline{x})(y-\overline{y}) = 700.64$	$\sum(x-\overline{x})^2 = 819.88$

Step 2: Calculate means:

$\overline{x} = 48.29$ and $\overline{y} = 51.57$

Step 3: Calculate slope and intercept:

$$\beta_1 = \frac{700.64}{819.88} = 0.855$$

$$\beta_0 = 51.57 - 0.855(48.29) = 10.27$$

Step 4: Line of regression

$$y = 0.855x + 10.27$$

Step 5: Estimate sales for $x = 30$

$$y = 0.855(30) + 10.27 = 36.21$$

Interpretation: The estimated sales correspond to an advertising expenditure of Rs.30 lakhs is 36.21 crores.

Q.4. From the data given below:

Marks in Economics (x)	25	28	35	32	31	36	29	38	34	32
Marks in Statistics (y)	43	46	49	41	36	32	31	30	33	39

Find the most likely marks in Statistics when the marks in Economics is 30.

Q.5. For a dataset with $n = 15$ points, the following information is given:

$\sum(x-\overline{x})(y-\overline{y}) = 80$

$\sum(x-\overline{x})^2 = 50$

$SS_{Total} = 150$

Calculate:

(a) The slope of the regression line.

(b) The regression sum of squares (SS_{reg}).

(c) The residual sum of squares (SS_{res}).

(d) R^2 value

Solution:

Step 1: Calculate slope of the regression line:

$$\beta_1 = \frac{\sum(x-\overline{x})(y-\overline{y})}{\sum(x-\overline{x})^2}$$

$$= \frac{80}{50} = 1.6$$

Step 2: Calculate SS_{reg}

$$SS_{reg} = \beta_1 . \sum(x-\overline{x})(y-\overline{y})$$

$$= 1.6\ (80) = 128$$

Step 3: Calculate SS_{res}

$$SS_{res} = SS_{Total} - SS_{reg}$$

$$= 150 - 128 = 22$$

Step 4: $R^2 = \frac{SS_{reg}}{SS_{Total}}$

$$= \frac{128}{150} = 0.855$$

Q.6. A dataset consists of 10 data points, with the following summary statistics provided:

$\sum(x-\overline{x})(y-\overline{y}) = 50$

$\sum(x-\overline{x})^2 = 40$ and $SS_{Total} = 100$

Use this information to answer the following questions:

(a) Find slope of the regression line.

(b) R^2 value

Q.7. A student's final score Y depends on:

X_1: Studying hours per week

X_2: Attendance percentage

The regression model is:

$$Y = 40 + 5X_1 + 0.2X_2$$

Predict the score of a student who studies 10 hours per week and has 80% attendance.

Solution: Substitute $X_1 = 10$ and $X_2 = 80$ in the given regression model equation

$$Y = 40 + 5(10) + 0.2(80)$$

$$Y = 106$$

Q.8 In a multiple regression model predicting life expectancy, the coefficients are:

$$Y = \beta_o + 0.4 * \log(income) - 0.2 * unemployment\ Rate + \varepsilon$$

Interpret the coefficient $\beta_1 = 0.4$ and $\beta_2 = -0.2$

Solution: 1. Interpreting $\beta_1 = 0.4$ for log $(income)$

- The coefficient 0.4 represents the change in Y (life expectancy) for a 1-unit increase in log $(income)$, holding all other variables constant. It means that a 1-unit increase in log $(income)$ corresponds to 0.4 years increase in life expectancy.

2. Interpreting $\beta_2 = -0.2$ for unemployment Rate

- The coefficient -0.2 represents the change in Y (life expectancy) for a 1-unit increase in unemployment Rate, holding all other variables constant. It indicates that for every 1% increase in the unemployment rate results in 0.2 -years decrease in life expectancy.

Q.9. In a multiple regression model predicting salary, the coefficients are:

$$Y = 6 + 0.5 * \log(Income) - 0.3 * Education\ years + \varepsilon$$

Interpret the coefficient $\beta_1 = 0.5$ for log $(Income)$ and $\beta_2 = -0.3$ for Education years

Solution: **1. Interpreting $\boldsymbol{\beta_1} = 0.5$ for log $(\boldsymbol{Income})$**

- The coefficient 0.5 represents the change in Y for a 1-unit increase in log $(Income)$, holding all other variables constant. It means that for every 1-unit increase in log $(Income)$,the salary increases by 0.5 lakhs.

2. **Interpreting $\boldsymbol{\beta_2} = -0.3$ for Education years**

- The coefficient -0.3 represents the change in Y for a 1-unit increase in unemployment Rate, holding all other variables constant. It indicates that for every 1-year increase in Education years, the salary decreases by 0.3 lakhs.

0.0.1 Example 1: Study Hours vs. Exam Score

Can we predict a student's final exam score based on their study hours? Consider the dataset:

Study Hours (x)	Exam Score (y)
2	65
4	75
6	85
8	90

Detailed Solution:

1. Calculate means:

$$\bar{x} = \frac{2+4+6+8}{4} = 5$$
$$\bar{y} = \frac{65+75+85+90}{4} = 78.75$$

2. Calculate required sums:

x_i	y_i	$(x_i - \bar{x})$	$(y_i - \bar{y})$	$(x_i - \bar{x})(y_i - \bar{y})$
2	65	-3	-13.75	41.25
4	75	-1	-3.75	3.75
6	85	1	6.25	6.25
8	90	3	11.25	33.75
			Total:	85

$$\sum (x_i - \bar{x})^2 = (-3)^2 + (-1)^2 + (1)^2 + (3)^2 = 20$$

3. Calculate slope:

$$\beta_1 = \frac{85}{20} = 4.25$$

4. Calculate intercept:

$$\beta_0 = 78.75 - 4.25(5) = 57.5$$

5. Final equation:

$$\hat{y} = 57.5 + 4.25x$$

6. Interpretation:

   - For each additional hour of study, the exam score is expected to increase by 4.25 points
   - A student who doesn't study at all (x = 0) is expected to score 57.5 points

0.0.2 Example 2: Investment Returns

Dataset: Annual investment amount (x, in thousands) vs. Returns (y, in thousands)

Investment (x)	Return (y)
10	1.2
15	1.8
20	2.3
25	3.0
30	3.5

Problem: Find the regression equation and predict the return for an investment of $22,000.

Solution:

1. Calculate means:
$$\bar{x} = 20, \quad \bar{y} = 2.36$$

2. Create computation table:

x_i	y_i	$(x_i - \bar{x})$	$(y_i - \bar{y})$	$(x_i - \bar{x})(y_i - \bar{y})$
10	1.2	-10	-1.16	11.6
15	1.8	-5	-0.56	2.8
20	2.3	0	-0.06	0
25	3.0	5	0.64	3.2
30	3.5	10	1.14	11.4
			Total:	29

$$\sum (x_i - \bar{x})^2 = 250$$

3. Calculate coefficients:
$$\beta_1 = \frac{29}{250} = 0.116$$
$$\beta_0 = 2.36 - 0.116(20) = 0.04$$

4. Regression equation:
$$\hat{y} = 0.04 + 0.116x$$

5. Prediction for x = 22:
$$\hat{y} = 0.04 + 0.116(22) = 2.592$$

6. Interpretation:
   - For each \$1,000 increase in investment, return increases by \$116
   - Expected return for \$22,000 investment is \$2,592

1 Multiple Linear Regression

Extension to multiple predictors:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + ... + \beta_p x_p + \epsilon$$

1.1 Model Assessment

1.1.1 R-squared

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}} = \frac{SS_{reg}}{SS_{tot}}$$

1.1.2 F-test

Testing overall model significance:

$$F = \frac{MS_{reg}}{MS_{res}}$$

1.2 Complex Example: House Price Prediction

Dataset with multiple features:

- Size (sq ft)
- Number of bedrooms
- Age of house
- Distance from city center

Model:

$$Price = \beta_0 + \beta_1(Size) + \beta_2(Bedrooms) + \beta_3(Age) + \beta_4(Distance)$$

2 Practice Problems

2.1 Easy Level

1. Calculate the regression line for:

$$(x, y) : (1, 2), (2, 4), (3, 6)$$

> **Hint**
>
> Use the basic formulas for slope and intercept

2. Interpret the slope in: $salary = 30000 + 5000(years_experience)$

> **Hint**
>
> Think about what one unit increase in x means

2.2 Medium Level

1. For a dataset with $n = 20$ points, $\sum(x_i - \bar{x})(y_i - \bar{y}) = 100$, $\sum(x_i - \bar{x})^2 = 25$, calculate:

   (a) The slope

   (b) R-squared if $SS_{tot} = 150$

> **Hint**
>
> Use the formulas for slope and R-squared directly

2. In a multiple regression model, interpret the coefficient when:

   - $\beta_1 = 0.5$ for log(income)
   - $\beta_2 = -0.3$ for education_years

> **Hint**
>
> Consider the units and transformations

2.3 Difficult Level

1. A dataset shows evidence of heteroscedasticity. Propose and justify:

   (a) How to detect it

(b) How to address it

(c) Impact on inference

> **Hint**
>
> Think about residual plots and transformations

2. For the model $y = \beta_0 + \beta_1 x + \beta_2 x^2$:

   (a) Is this still linear regression? Why?

   (b) How to find the x-value that maximizes y?

> **Hint**
>
> Consider the parameters vs. variables distinction